

Update Doc 31/07/2026

I believe there was quite a few things (alot that claude missed from last time so will write again)

Mathematical Question Authoring and Import Tools

The platform should include a built in **mathematical equation editor** so teachers can create and edit questions using correctly formatted mathematical notation.

The editor should support:

1. Fractions, powers, roots and brackets
2. Algebraic expressions and equations
3. Inequalities
4. Indices and logarithms
5. Trigonometric notation
6. Calculus notation
7. Vectors, matrices and coordinate notation
8. Greek letters and common mathematical symbols
9. Superscripts and subscripts
10. LaTeX input and a clickable symbol palette
11. A live preview showing exactly how the mathematics will appear to students
12. Mathematical notation within questions, answer choices, hints, feedback and worked solutions

Teachers should be able to enter mathematics either by selecting symbols from a toolbar or by typing LaTeX. The system should store the original mathematical markup and render it consistently across desktop, tablet, mobile and printable materials.

Question Import Function

The platform should also include an **Import Question** function that allows teachers to upload existing questions from:

1. PDF files
2. Word documents
3. Images or screenshots
4. Scanned worksheets
5. Existing exam papers

Imported questions may contain:

1. Tables

2. Graphs and charts
3. Diagrams
4. Geometric figures
5. Mathematical notation
6. Images
7. Multiple question parts
8. Marks and answer spaces

When a question is imported, the system should extract and separate its content into editable components, including:

1. Question text
2. Mathematical expressions
3. Tables
4. Charts
5. Figures and diagrams
6. Answer options
7. Question parts
8. Marks available
9. Instructions
10. Source information

The teacher must be able to review and edit every extracted component before saving or publishing the question.

Edexcel Style Conversion

After import, the system should reformat the question into a clean and consistent **Edexcel inspired exam style**.

The purpose is to create a professional UK examination appearance while avoiding direct copying of Pearson Edexcel branding, logos, protected templates or copyrighted design elements.

The converted question should use:

1. Clear exam style typography
2. Consistent spacing and alignment
3. Numbered questions and subquestions
4. Marks displayed clearly, such as (2 marks)
5. Properly formatted mathematical notation
6. Clearly labelled tables, graphs and diagrams
7. Suitable working space where required
8. Command words such as *calculate*, *show that*, *explain* and *prove*
9. A clean black and white printable layout
10. Consistent margins and page breaks

Tables, graphs and figures should be preserved accurately. They should not be removed, simplified or changed in a way that alters the mathematical meaning of the question.

Where possible, imported charts and diagrams should be recreated as editable, high quality elements rather than being stored only as low resolution images.

Import Review Screen

Before adding an imported question to the question bank, show a review screen containing:

1. The original uploaded question
2. The converted exam style version
3. Editable question text
4. Editable mathematical notation
5. Editable marks
6. Editable tables and labels
7. Replaceable or adjustable figures
8. Learning objective selection
9. Difficulty selection
10. Correct answer and worked solution fields
11. Misconception based distractor fields
12. A preview of the student view
13. A preview of the printable version

Quiz (High Priority)

Adaptive Diagnostic Quiz System

1. Project Purpose

Build an adaptive diagnostic quiz system for students.

The purpose of the quiz is not only to identify whether an answer is correct or incorrect. It should identify the specific misconception behind each incorrect answer and provide personalised support until the student demonstrates understanding.

The quiz should adapt to each student in real time.

Each quiz may contain a different number of main questions depending on the topic.

For example, a short topic may contain 5 main questions, while a larger topic may contain 15 or more.

The number of questions should therefore be configurable.

2. Question Structure

Each main question must contain four answer options.

1. One correct answer
2. Three incorrect answers
3. Each incorrect answer must represent a specific misconception
4. Every misconception must have a unique misconception ID

Each question should include the following data:

1. Question ID
2. Topic ID
3. Learning objective ID
4. Difficulty level
5. Question text
6. Four answer options
7. Correct answer
8. Misconception ID linked to each incorrect answer
9. Feedback for the correct answer
10. Feedback for each misconception
11. Scaffold level
12. Question type
13. Explanation or worked solution

3. Initial Quiz Behaviour

The quiz should begin with easier questions and gradually move

towards more difficult questions.

Suggested progression:

1. Easy
2. Medium
3. Hard

The system should adapt the difficulty based on student performance.

If the student answers correctly without support, the next question may become more difficult.

If the student answers incorrectly, the system should remain on the same learning objective and provide more support.

4. Correct Answer Behaviour

When a student selects the correct answer:

1. Show positive feedback
2. Explain briefly why the answer is correct
3. Add energy to the energy meter
4. Update the student's mastery score
5. Save the result
6. Move to the next appropriate question

A correct answer on the first attempt should contribute more towards mastery than a correct answer reached after several scaffolded attempts.

The system should record:

1. Whether the answer was correct on the first attempt
2. Number of attempts
3. Number of scaffold levels used

4. Time spent
5. Whether hints were used
6. Difficulty level
7. Energy earned
8. Mastery gained

5. Incorrect Answer Behaviour

When a student selects an incorrect answer:

1. Identify the misconception connected to that answer
2. Save the misconception to the student profile
3. Display feedback that directly addresses the misconception
4. Do not move to the next main learning objective
5. Generate or select another question targeting the same misconception
6. Increase the level of scaffolding
7. Continue until the student answers correctly

Incorrect answers should not simply display “Try again”.

Example feedback:

“You worked from left to right without completing the division first. Division must be completed before addition and subtraction. Let’s break this down into smaller steps.”

The student should receive unlimited attempts.

6. Scaffolded Question System

Every time the student answers incorrectly, the next question should include more support.

The scaffolded question must continue targeting the same

misconception.

The distractors should also remain misconception based.

The system should not replace meaningful distractors with random incorrect answers.

Suggested scaffold sequence:

Scaffold Level 0

Original diagnostic question.

No hints.

Scaffold Level 1

A similar question with simpler numbers.

Include a short reminder of the relevant rule.

Scaffold Level 2

Break the question into smaller steps.

For example:

“Which calculation should be completed first?”

Scaffold Level 3

Provide a partially completed worked example.

For example:

$$20 - 8 \div 2 + 3$$

First calculate:

$$8 \div 2 = \underline{\quad}$$

Scaffold Level 4

Provide stronger visual support or guided choices.

For example:

1. Complete the division
2. Rewrite the expression
3. Work from left to right

Scaffold Level 5 and Beyond

Continue generating increasingly guided questions until the student answers correctly.

There should be no maximum number of retries.

The scaffolding should never give the answer immediately unless the system is showing a worked example before asking a new question.

7. AI Generated Questions

Questions should be generated by AI using controlled templates and validation rules.

The AI must generate:

1. A question
2. One correct answer
3. Three misconception based distractors
4. A misconception ID for each distractor
5. Correct answer feedback
6. Misconception specific feedback
7. A worked solution

8. Difficulty level
9. Learning objective
10. Scaffold level

The AI must not generate distractors randomly.

Each incorrect answer must be mathematically plausible and connected to a clearly defined error in reasoning.

The system should validate AI generated questions before showing them to students.

Validation should check:

1. There is exactly one correct answer
2. The correct answer is mathematically accurate
3. All four answers are different
4. Each incorrect answer maps to a misconception
5. The question matches the requested difficulty
6. The question matches the learning objective
7. The scaffold level is appropriate
8. The explanation is correct
9. No answer is accidentally duplicated
10. No distractor is also mathematically valid

Do not rely entirely on AI output without validation.

Use structured JSON responses from the AI.

8. Review Section

At the end of the main quiz, create a Duolingo style review section.

The review section should include learning objectives or misconceptions where the student did not answer correctly on the

first attempt.

The review question should:

1. Assess the same concept
2. Target the same misconception
3. Use different numbers or wording
4. Avoid repeating the exact original question
5. Be appropriate to the student's current scaffold level
6. Contribute towards the final quiz score

Students must answer each review question.

If they answer a review question incorrectly, the adaptive scaffolding process should continue until they answer correctly.

The review section is part of the overall assessment and should affect the student's final percentage.

9. Pass Rate

The pass mark is 80 percent.

The final score should include both:

1. Main quiz performance
2. Review section performance

The score should not be based only on whether the student eventually reaches the correct answer.

The scoring system should take account of:

1. First attempt accuracy
2. Number of incorrect attempts
3. Amount of scaffolding required

4. Review performance
5. Difficulty of questions
6. Mastery demonstrated

If the student finishes the main quiz and review section with less than 80 percent:

1. Mark the quiz as not yet passed
2. Ask the student to complete the quiz again
3. Generate a new version of the quiz
4. Use similar styled questions
5. Focus more heavily on the misconceptions previously identified
6. Do not repeat the exact same questions unless necessary

10. Quiz Retake Behaviour

Every retake should generate a different quiz.

The new quiz should:

1. Cover the same required learning objectives
2. Use different values, wording or contexts
3. Include additional questions targeting unresolved misconceptions
4. Begin at an appropriate level based on previous performance
5. Continue to use easy to hard progression
6. Avoid repeating exact question combinations

The system should store all previous question IDs to reduce repetition.

11. Misconception Memory Across Topics

If a student answers a misconception based question incorrectly more than once, that misconception should be marked as persistent.

A persistent misconception should be included in future quizzes, even when the student moves to another topic, where mathematically relevant.

For example, if a student repeatedly struggles with order of operations, suitable order of operations checks may appear within future quizzes on algebra, fractions or substitution.

The misconception should remain active until the student demonstrates mastery.

Once the student consistently answers questions related to that misconception correctly, it should be removed from the active misconception list.

Suggested rule:

1. One incorrect answer means the misconception is identified
2. More than one incorrect answer means the misconception is persistent
3. A persistent misconception is added to future relevant quizzes
4. The misconception is removed after the student demonstrates secure understanding

The system should only insert the misconception into future topics when it is genuinely relevant.

12. Mastery Levels

Each learning objective should have one of four mastery levels.

Not Started

The student has not attempted the learning objective.

Learning

The student has attempted the objective but required significant support or made repeated errors.

Developing

The student has demonstrated some understanding but may still require occasional support.

Mastered

The student can answer suitable questions correctly with little or no support.

Suggested mastery score ranges:

Not Started: 0

Learning: 1 to 39

Developing: 40 to 79

Mastered: 80 to 100

Mastery should be stored separately for each learning objective.

Misconception mastery should also be stored where appropriate.

13. Suggested Mastery Scoring

Use a configurable scoring model.

Example:

Correct on first attempt: +20 mastery points

Correct after one scaffold: +12 mastery points

Correct after two scaffolds: +8 mastery points

Correct after three or more scaffolds: +5 mastery points

Correct review question on first attempt: +10 mastery points

Incorrect response: no mastery increase

Repeated misconception: reduce confidence score or flag for future practice

Do not reduce mastery so aggressively that students feel punished.

The emphasis should be on demonstrated learning and improvement.

14. Energy Meter

Display an energy meter in the top right corner throughout the quiz.

The energy meter should act as positive reinforcement.

Suggested energy rules:

Correct on first attempt: +15 energy

Correct after one scaffold: +10 energy

Correct after two or more scaffolds: +5 energy

Correct review question: +5 energy

Completing a learning objective: bonus energy

Mastering a learning objective: bonus energy

Incorrect answers should not remove energy.

The energy meter should:

1. Animate when energy is gained
2. Display the amount gained
3. Show current energy as a percentage or value
4. Fill visually like a battery
5. Trigger a celebration when full
6. Reset or continue according to quiz settings

Example:

 +15 Energy

Energy: 65 percent

15. Student Interface

The student interface should display:

1. Quiz title
2. Topic
3. Current question
4. Question number
5. Progress bar
6. Energy meter in the top right
7. Four answer buttons
8. Feedback area
9. Hint area when appropriate
10. Mastery update after important milestones

The design should feel encouraging, modern and game based.

It should take inspiration from Duolingo, Khan Academy and similar mastery based learning platforms without directly copying their branding.

16. Progress Bar Behaviour

The progress bar should represent progress through the current quiz.

Scaffolded questions should not necessarily increase the main question number.

For example:

Main Question 4 of 10

Support Step 2

The student should understand that they are still working on the same learning objective.

The progress bar should not move backwards.

17. Teacher Dashboard

Include a teacher dashboard showing:

1. Student name or ID
2. Quiz completion status
3. Final score
4. Pass or not yet passed
5. Number of attempts
6. First attempt accuracy
7. Persistent misconceptions
8. Current mastery level by learning objective
9. Number of scaffolds required
10. Review performance
11. Energy earned
12. Time spent
13. Questions answered
14. Topics requiring additional support

Teachers should be able to see the difference between:

1. A student who answered correctly immediately
2. A student who answered correctly after several scaffolded attempts

19. Question Selection Logic

Use this priority order when choosing the next question:

1. Continue unresolved current misconception
2. Increase scaffolding if the previous answer was incorrect
3. Continue until the current learning objective is answered correctly
4. Move to the next main learning objective
5. Increase difficulty when the student is succeeding
6. Include persistent misconceptions when relevant
7. Complete the main quiz
8. Start the review section
9. Calculate final score and mastery
10. Pass the student if the score is at least 80 percent
11. Require a new quiz attempt if the score is below 80 percent

Suggested pseudocode:

start quiz

for each learning objective:

 present diagnostic question

 while answer is incorrect:

 identify misconception

 save misconception

 increase scaffold level

- generate scaffolded question
- present scaffolded question

- update mastery
- add energy
- move to next learning objective

generate review questions for all objectives not correct on first attempt

for each review objective:
 present a new similar question

- while answer is incorrect:
 - identify misconception
 - increase scaffold level
 - present another scaffolded question

calculate final score

if score ≥ 80 :
 mark quiz as passed
else:
 create a new quiz attempt
 prioritise unresolved misconceptions

20. Technical Requirements

Build the system using modular components.

Suggested components:

1. QuizEngine
2. QuestionGenerator
3. QuestionValidator
4. MisconceptionTracker
5. ScaffoldManager
6. MasteryManager
7. EnergyManager
8. ReviewManager
9. ScoreCalculator
10. StudentProgressStore
11. TeacherDashboard
12. AIService

Keep the quiz logic separate from the user interface.

Use configuration files for:

1. Pass rate
2. Energy rewards
3. Mastery rewards
4. Number of questions
5. Difficulty progression
6. Maximum energy
7. Scaffold types
8. Review rules

Do not hard code these values into multiple components.

21. Important Rules

1. There must always be exactly one correct answer
2. Incorrect answers must represent real misconceptions
3. Students must continue until they answer correctly

4. Scaffolding must increase after every incorrect response
5. Students have unlimited retries
6. Review questions contribute to the final score
7. Students must achieve at least 80 percent to pass
8. Students who do not pass must complete a new quiz
9. Retake questions must be different but assess the same objectives
10. Persistent misconceptions must appear in future relevant quizzes
11. Persistent misconceptions must be removed when securely resolved
12. Correct answers increase energy
13. Incorrect answers do not remove energy
14. Mastery must be tracked by learning objective
15. AI generated questions must be validated before use

22. First Development Stage

For the first version, build a working prototype using one mathematics topic.

Suggested prototype topic:

Order of operations

Include:

1. Three learning objectives
2. At least five question templates per learning objective
3. Three misconception based distractors per question
4. Unlimited adaptive scaffolding
5. Review section
6. 80 percent pass rule

7. Energy meter
8. Mastery levels
9. Local storage or a simple database
10. Basic teacher results page

Once the prototype is working correctly, make the system reusable for additional topics.

Additionally display the difficulty in terms of chillies mild, spicy , extra hot instead of easy median and hard.